



Southern California Edison
CS Re-Platform Project

CIM-CS-07 - Handle Bill Inquiries

Customer Interaction Management

Business Process Design

Version 0.10

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1 Business Process Overview

The Handle Bill Inquiry process defines how to manage a variety of questions from a Business Partner (BP) regarding their bill, which will be done via Hybris C4C. These topics can range from general billing questions to E-Bill requests or Energy Advice Requests. If the inquiry is resolved by the CCC or BCD representative during the call with the BP, an Activity will be manually created by the CCC or BCD representative to document the call and no further action is required. However, if the inquiry is not resolved by the CCC or BCD representative, an Activity will be created stating such and a ticket or a BPEM case will be created and routed to the appropriate workgroup for further handling.

Current Pain Points for Southern California Edison

- General Bill Inquiries
 - There are times when the BI Rep needs to forward the BI to additional depts. for further research.
 - Not all agents are able to address bill inquiries from customers, in some cases BI's are being created when perhaps they could be resolved with the first call.
 - No visibility to the status of Billing Inquiry
- E-Bill
 - Not a consistent approach across all channels when offering Paperless billing when the customer has already declined.
 - Customer does not have access to historical bills in online following registration in Paperless Billing (only subsequent bills will appear). Results in call into the call center and confusion why they don't have access to their data.
 - Separate Customer Accounts for Deposit and Energy results in an incomplete view of the charges to the customer on their paperless bill which includes only Energy charges.
 - Auto enrolling / defaulting customers to Paperless Billing when they make an online payment is resulting in calls into the call center / complaints
 - Paperless bills on My Account can only be associated to one user.
 - From the CCC, agents are not viewing the bill in the same format making referencing the bill with the customer over the phone difficult.

Responsible Parties

Contact	Contact Person
Southern California Edison Business Process Owner	Janet Covington
Southern California Edison Lead	Christine Evans
HCL Functional Lead	Adrianus Halim
Solution Architect	Pedro Costa
Front Office Lead	Christine Evans

Contact	Contact Person
Subject Matter Experts	Deborah Tillman Scott Hirth
Key User Group	CCC, BCD, RSO

Document Control

Version	Description of Change	Date	Author
0.1	Creation of Initial Version	01/08/2018	Jessica Marsillo
0.2	Review	03/01/2018	Tuan Howe Yap
0.3	Update	03/08/2018	Jessica M / Tuan Y
0.4	Update based on Adrian review	03/11/2018	Tuan Howe Yap
0.5	Review and update	03/11/2018	Jessica Marsillo
0.6	Update based on Pedro review	03/12/2018	Jessica M/ Tuan Yap
0.7	Updated Visio	03/13/2018	Jessica Marsillo
0.8	Update based on CSRP review	03/15/2018	Tuan Howe Yap
0.9	Update based on Cari review	03/19/2018	Tuan Howe Yap
0.10	Updated based on Cari review	03/20/2018	Tuan Howe Yap
0.11	Update based on business review	03/22/2018	Tuan Howe Yap

Terminology

Term/Acronym	Definition and SAP Equivalent (if applicable)
APSE	Accelerated (or Axon) Project Support Environment
BDEX	SAP add-on functionality to support the processing of BPEM Cases
Billing Cycle	Is a group of contracts to be billed together via date-driven or regional criteria, it controls the billing cycle and is linked to the Meter Reading Unit. SAP equivalent term is the Portion.
Billing Document	A document that outlines the charges incurred for energy consumption for a specific contract. During the process of Billing, a billing document is generated for each contract on a contract account.
BP Contact	This can also be called as Customer Contact in S/4HANA acting as an interaction log to record activities related to a customer
BMD	Business Master Data. The customer information which includes personal details, payment terms, dunning levels, billing rates and rate categories. The business master data objects are Business Partner, Contract Account and Contract.
BWR	Basic Work Request. SAP equivalent is BPEM.

Term/Acronym	Definition and SAP Equivalent (if applicable)
CA	Contract Account
Channel	Data object in SAP that holds the interval data. Intervals can be kwh, price, factor, etc. SAP equivalent is Profile.
Clarification	A list of payments that could not be processed and require additional work in order to be posted.
Connection Object	Link between installation and postal regional structure. Usually a building, but can also be a piece of property or other facility, such as a fountain or construction site. The connection object corresponds to the functional location in plant maintenance (PM).
Customer	Any entity with a relationship with SCE. This term is usually used in reference to customers (both individuals and organizations) This can include employees and third party suppliers (banks and payment partners). SAP Equivalent is Business Partner (BP)
BPD	Business Process Design (or Document)
BPEM	Business Process Exception Management
EAM	SAP Enterprise Asset Management
Hybris C4C	SAP Cloud for Customer Solution
IMG	SAP Implementation Guide - where configuration is performed
Level Pay Plans	SAP equivalent term is Budget Billing. Budget Billing allows customers to pay regular equal amounts over a yearly period.
S/4HANA	SAP HANA 4 Business Suite
SCE	Southern California Edison
Summary Billing	Standard SAP functionality that enables the invoicing of multiple Contract Accounts under one Parent Contract Account (Consolidated Summary Billing). SAP equivalent is Collective Billing.
TMD	SAP Technical Master Data e.g. Connection Object, Premise and Device Location

Additional terms may be found in APSE [General Document # 32 CSRP Glossary](#)

2 Business Process Design

User will always authenticate and validate that the requestor is an authorized user before providing account information or carry out transaction for the requestor.

Handle Bill Inquiry Details

A BP may contact a CCC or BCD representative with a variety of questions regarding their bill, which can be answered using various screens in both Hybris C4C and S/4HANA.

Billing Overview and Bill Images

The Billing Overview Screen in Hybris C4C provides a single view of the billing/invoice history of the Business Partner, including 36 months of billing history. Here, the CCC or BCD Representative can find the invoice date, amount due, date open amount, payments received, and whether or not a bill reversal has occurred (Figure 1: Billing Overview).

Invoice Payments Open Items Installment Plan Payment Plan Premise Contracts with Meter Reading							
Invoices (9)							
Invoice No	Invoice Date	Amount Due	Due Date	Open Amount	Payments	Reversed	
60032130	03/01/2017	437.22 USD	03/15/2017	387.49 USD	49.73 USD	✓	
60032131	03/22/2017	447.85 USD	04/05/2017	398.12 USD	49.73 USD	✓	
60032132	03/22/2017	731.39 USD	04/05/2017	294.17 USD	437.22 USD	✓	
60032133	03/22/2017	750.57 USD	04/05/2017	313.35 USD	437.22 USD	✓	
60032134	03/22/2017	793.42 USD	04/05/2017	793.42 USD	0.00 USD	✓	

Figure 1: Billing Overview

From the Billing Overview Screen in Hybris C4C, the CCC or BCD Representative can access the associated Bill Images by clicking on the Invoice Number hyperlink (Figure 2: Bill Image). Using the Bill Image, the CCC or BCD representative can view and compare bills in multiple sessions within Hybris C4C with the following.

Alternatively, user can navigate to Consolidated Bill view mentioned on the next page.

- Bill Period
- Amount Due
- KWH
- KW
- KVAR
- Usage by Consumption Period (On Peak, Off Peak, etc.)
- Period Days
- Credit and debits based on demand response event participation, regulatory adjustment (i.e. climate credit, ERRa), rebates or miscellaneous adjustments
- NEM Settlement Bills (used to review monthly information)

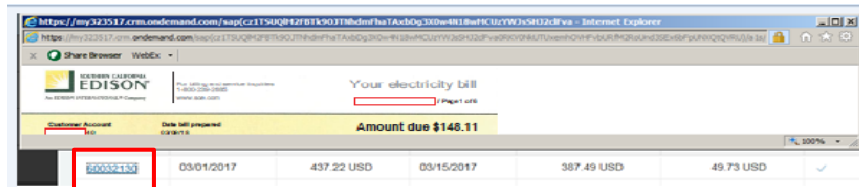


Figure 2: Bill Image

Using T-Code ES32 in S/4HANA or via FOP (*E-CIM-CS-02: Transaction Launcher*) in Hybris C4C, the CCC or BCD Representative can view the billing periods and determine if invoicing is taking place by clicking on the “Billing Periods” button (Figure 3: View Billing Periods).

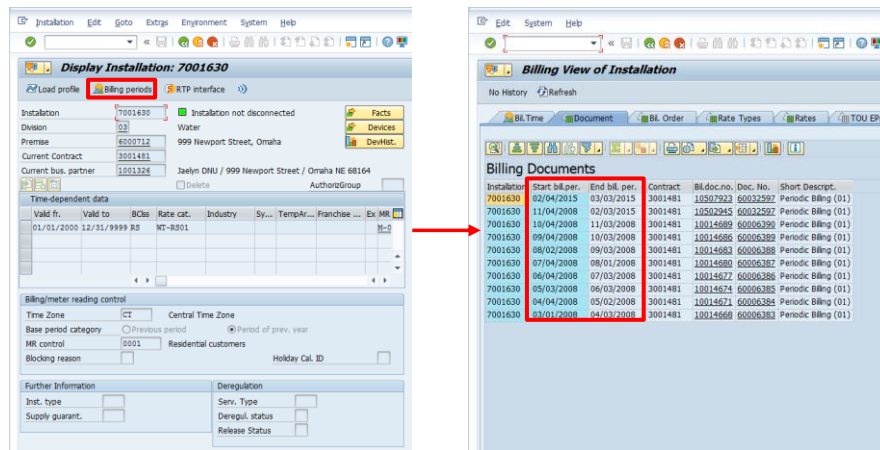


Figure 3: View Billing Periods

To view whether a contract is NEM (Net Energy Metering), VNM (Virtual Net Metering), MASH, etc., the CCC or BCD Representative should view the Installation Facts via Hybris C4C Billing Parameters section (Figure 4: Billing Parameters). Representative can use the filter to select the type of Installation Facts that he/she wanted to see.

Home

TOU-D-T SCE02

TOU-D-T SCE02

TOU-D-T SCE02

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UNT TEAM

INSTALLATION POINTS

ACTIVITIES

UTILITIES SERVICE NOTIFIC...

UTILITIES SALES POD

UTILITIES RATE CHANGE

🔖

Contracts (1)

🔍

🔗

📄

⬇️

Refresh

Rate Change

Contract Account External ID	Contract ID	Service Address	Installation ID	Division	Move-In Date
25063602	3011438	110 Alcovy Road San Diego...	7012145	Electric	01/01/2016

🔖

Billing Parameters (2)

🔗

🔍

Operand	Operand Category
A_BILLED	Amount
A_BILLED	Amount

+

Web Services

Actions

Summary

Move In

Move Out

Transfer

Hierarchy

Create Contract Account

Laun...

Figure 4: Billing Parameters

Installation Facts is also used to host other data, such as Usage data, Billing History, Program & Product etc. This information will be available in the Billing Parameters view in Hybris C4C.

Duplicate Bill

If a Business Partner (BP) requests a duplicate bill to be sent to them based on their preferred communication method, a CCC or BCD representative can do so via OpenText (Refer to BPD [CIM-CS-09 - Manage Outbound Request](#) for more information) or refer the Business Partner (BP) to sce.com to reprint the statement(s).

Consolidated Bill View

BP may request a summary of past meter reading and billing history while on the phone with user, user will navigate to a Hybris C4C FOP (tcode ZCon_Bill_View) to view the information in Consolidated Bill View (Figure 5: Consolidated Bill view). By using this view, user does not need to open up multiple bill images to obtain the information. This view contains the followings information:

- Meter Reading Date
- KWH
- KW
- Billing Days
- Total KWH Usage
- Avg kWh Usage per day
- Daily Cost
- Bill Amount

This customized screen will be made available in S/4HANA for optimization purpose and will be done using enhancement *E-CIM-CS-17 - Consolidated Bill View*.

Ending Meter Read Date	KWH Read	KW Demand	Billing Days	Total KWH Usage	Avg KWH Usage per Day	Daily Cost	Total Service Account Charges
10/02/17	0	0.00	32	1,110	34.68	\$2.88	\$17.40
09/01/17	0	0.00	29	1,544	53.24	\$6.63	\$192.41
08/03/17	0	0.00	29	1,746	60.21	\$7.80	\$226.10
07/05/17	0	0.00	30	1,411	47.03	\$5.60	\$167.96
06/05/17	0	0.00	32	821	25.66	\$3.62	\$115.99
05/04/17	0	0.00	30	590	19.93	\$2.89	\$86.84
04/04/17	0	0.00	29	500	17.52	\$1.42	\$41.22
03/06/17	0	0.00	32	335	10.47	\$1.31	\$41.93
02/02/17	0	0.00	29	277	9.55	\$1.16	\$33.52
01/04/17	0	0.00	30	276	9.20	\$1.03	\$30.90
12/05/16	0	0.00	33	432	13.09	\$1.64	\$54.00
11/02/16	0	0.00	30	578	19.27	\$4.05	\$121.40
10/03/16	0	0.00	32	1,015	31.72	\$3.83	\$122.61
09/01/16	0	0.00	29	1,497	51.62	\$8.99	\$260.77
08/03/16	0	0.00	29	1,969	67.90	\$12.72	\$368.97
07/05/16	0	0.00	32	2,369	74.03	\$14.13	\$452.13
06/03/16	0	0.00	30	1,091	36.37	\$8.74	\$262.33
05/04/16	0	0.00	29	641	22.10	\$4.91	\$142.38
04/05/16	0	0.00	32	604	18.88	\$2.75	\$87.95
03/04/16	0	0.00	30	462	15.40	\$2.94	\$88.21
02/03/16	0	0.00	29	285	9.83	\$1.61	\$46.76

Figure 5: Consolidated Bill view

NEM Inquiry

Customer may call with questions related to their NEM bill. To provide information to customer, user will navigate to OpenText xECM (Enterprise Content Management) to view the NEM bill image or in Hybris C4C navigate to the invoice view and click on the hyperlink of the invoice number. Refer to BPD [CIM-CS-09 Handle Inbound Request](#) for more information.

For information on NEM Tracking History, FOP will be used to show these information. Refer to BPD [UB-BI-12 Calculate Energy Bill – Net Metering](#).

Rate Comparison and Analysis

A rate comparison for current and past rates can be viewed using Hybris C4C. This feature will be released in the SAP C4C release 1805. The rate comparison in Hybris C4C allows CCC or BCD representative to select BP existing rate and select with a different rate for comparison. This provides the representative the dollar amount comparison based consumption.

Additionally, a CCC or BCD representative can complete a rate analysis via SCE.COM by entering Contract number and postal code. Super users will be able to view rate analysis with more advanced features in GridX outside of SAP. The GridX will then be launched and the CCC/BCD representative can log in using their assigned username. If the CCC or BCD representative is unable to obtain the rate analysis via SCE.COM or GridX, they will create a Ticket with the category 'Rate Analysis Request'. The Ticket should be routed to the CCC Offline Support to or BCD Rate Analysis COE depending on the customer type.

User Utility Tax (UUT)

UUT is a tax that is imposed by certain cities for electric service. To view if the contract is subject to UUT, CCC or BCD representative will navigate to Installation screen (tcode ES32) via FOP in Hybris C4C. Entry populated in Franchise Contract indicates contract has UUT (Figure 6: Franchise Contract).

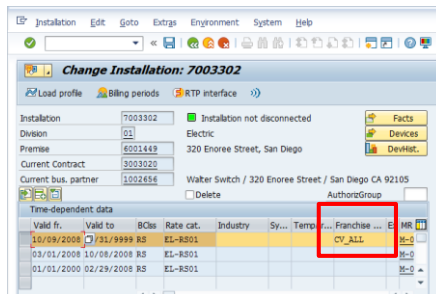


Figure 6: Franchise Contract

Should a Business Partner (BP) request UUT exemption, the representative can create a BPEM using category “UUT exemption”. If customer refuses to pay the UUT charges, the system will not be disconnected due to non payment. The representative can create a BPEM using category “UUT Refused to Pay”. These BPEM cases go to RSO Billing.

Bill Adjustments/Reversal/Rebill

If a front office employee believes a bill correction needs to be completed, the C4C user will create a Ticket to the CCC BI team. The BI Team will confirm if a reversal/rebill/correction to bill is needed. If the meter is a mechanical (Register) meter, the BI Team will complete the reversal and correct usage as needed. If the meter is an IDR meter, the BI team will create a BPEM case with the category ‘Bill Correction’, to RSO Billing. All other C4C users who believe a bill correction needs to be completed by back office will create a BPEM case with the category ‘Bill Correction’, to RSO Billing.

The **Full Reversal**, **Adjustment Reversal** and **Create Invoice** buttons in Hybris C4C will be hidden. The **Full Reversal** will be available in Hybris C4C for certain groups (e.g., BI Team).

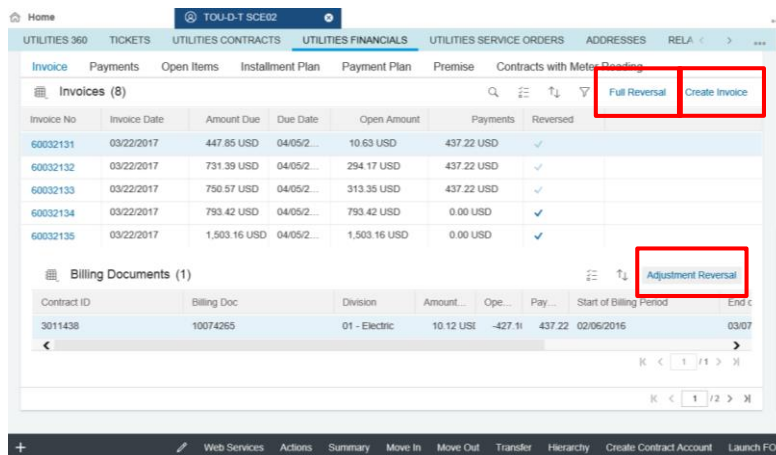


Figure 7: Bill Adjustment

Service Orders

When necessary, the CCC or BCD representative will be able to create a Service Order to resolve a Business Partner's inquiry through Hybris C4C. Refer to BPD [CIM-CS-05 - Handle Request - Process Field Work Request](#) for more information.

E-Bill

E-Bill allows a BP to receive their bills electronically. The BP will receive an email notification from SCE containing the hyperlink that directs them to SCE.com when an electronic copy of their bill is ready. Up to 20 months of billing will be available for viewing on SCE.com at the time of SAP go-live.

A CCC or BCD representative can view the BP's enrollment in E-Bill on the Contract Account screen (tcode CAA3) in Hybris C4C via FOP (*E-CIM-CS-02: Transaction Launcher*).

The representative can also determine how the BP was enrolled in E-Bill (example: CheckFree, SCE.com or IVR) by viewing the Change History Log in Contract Account (Figure 7: CA Change History) via FOP in Hybris C4C (t-code CAA3). A CCC or BCD representative can be identified by their user IDs, while a unique user ID will be created for CheckFree/SCE.COM/IVR for easy identification of when a BP enrolls in E-Bill via these methods.

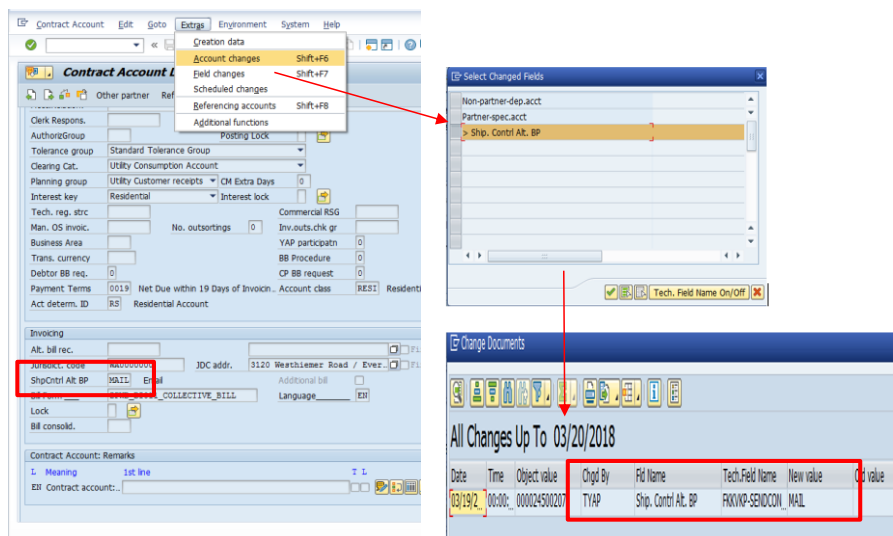


Figure 7: CA Change History

Enrollment

Requests to enroll in E-Bill are received by phone, SCE.com, IVR, marketing campaign, online payment default, credit card (applicable for residential customers with MyAccount) or third-party (Check Free). Refer to BPD [CIM-CS-10: Manage Customer Self-Service](#) for more information about E-Bill enrollment and MyAccount.

Should more than one BP wish to receive/enroll in E-Bill for the same Contract Account, the CCC or BCD representative will use the 'Alt. Bill Rec.' field on the Contract Account screen via FOP (*E-CIM-CS-02: Transaction Launcher*), to send invoices to multiple Business Partners (BPs).

Once a BP agrees to enroll in E-Bill, the CCC or BCD representative will use FETCH (if needed) to read the Terms and Conditions of E-Bill to the BP. The CCC or BCD representative will then manually create an Activity to capture the acceptance of the Terms and Conditions by the BP.

De-enrollment from E-Bill

Requests to de-enroll in E-Bill are received by phone, e-mail, in person, CheckFree, SCE.com and IVR. A CCC or BCD representative can de-enroll a BP's contract account(s) in E-Bill on the Contract Account screen in Hybris C4C. There will be no changes to the de-enrollment process via SCE.com or IVR.

Once the de-enrollment process is complete, the CCC or BCD representative will create an Activity in Hybris C4C manually to capture the reason for de-enrollment from E-Bill.

Collective/ Joint Invoicing

Collective and Joint Invoicing is also known as Summary Billing in legacy CSS. When a BP requests Collective Invoicing or wishes to modify their Collective accounts, user will create a BPEM Case Category 'Collective Billing Request/Inquiry' from a Ticket via Hybris C4C manually. BPEM case for Collective Billing is document in BPD [CIM-CS-02 - Handle Customer Move-in/Out - Contract New](#). For Collective Invoicing, the child accounts must be manually linked to the master Contract Account by the back-office.

For Joint Invoicing, the representative will navigate to the S/4HANA Contract screen (t-code ES21) via Hybris C4C FOP (*E-CIM-CS-02: Transaction Launcher*). Refer to BPD [UB-BI-10: Produce Invoice - Collective and Joint Invoicing](#) for steps in setting up Collective and Joint Invoicing.

Budget Billing Plan

Budget Billing Plan is also known as Level Payment Plan in legacy CSS. The CCC or BCD representative will create a Budget Billing Plan when the BP requests to pay their annual bill amount in a fixed bill amount across a 12 month period. The representative will use the 'Create Payment Plan' function in Hybris C4C to create the Budget Billing Plan (Figure 8: Budget Billing Plan). To enroll a BP on Budget Billing Plan, customer has to meet eligibility requirement defined. Refer to BPD [UB-BI-11: Manage Billing Arrangements - Payment Plan](#) for more information about the Budget Billing Plan.

If needed, the user will be able to create a Ticket category 'Budget Billing Plan' to Offline Line Support or the BCD COE to setup or modify Budget Billing for a BP.

Contract ID	Move-In Date	Move-Out Date
3500186	06/28/2016	Unlimited
3500188	06/27/2016	Unlimited
3500187	06/27/2016	Unlimited

Payment Plan Parameters

Payment Plan Type
Payment plan category: BBP
Start Month: February
Start Year: 2018
Alternate Start Month: March

Figure8: Budget Billing Plan

Meter Readings

A Business Partner (BP) may inquire about meter readings, such as questions about estimates and actual meter readings.

1. If meter is a non-smart meter or also known as Register Meter, the representative will navigate to Hybris C4C Meter Reading view. (Figure 9: C4C Meter Reading). Meter readings for the 50 thousand non smart meters will be available here.
2. If the meter is a smart meter, the representative will navigate to S/4HANA Installation screen (t-code ES32) via Hybris C4C FOP (*E-CIM-CS-02: Transaction Launcher*). The billing determinants for 4.5 million meters will be available here via EDM (Energy Data Management). Interval Data is viewable using t-code EEDM02) via Hybris C4C FOP (*E-CIM-CS-02: Transaction Launcher*).
3. If the representative requires the back office to correct a meter reading or obtain meter reading information, they will create a BPEM case from a Ticket with the category 'Bill Correction', which will be routed to RSO Billing.

Figure9: C4C Meter Reading

Dunning & Payment Lock

Should a BP choose to dispute a bill, a Dunning Lock can be used to hold the amount of the bill being disputed with a start date and end date. By placing a Dunning Lock, the BP will be excluded from Collections (Dunning). Furthermore, a CCC or BCD representative may place an Incoming Payment Lock in Hybris C4C Open Item view to put a hold on any Direct Payments that may occur during the bill investigation with a start date and an end date (Figure 10: Create Dunning/Payment Lock). The creation of Dunning Lock and Incoming Payment Locks in both S/4HANA and Hybris C4C will be covered in the BPD [CC-FI-04: Manage Collections \(Dunning\) - Manage Credit and Collections](#). The user should also add an Activity record on the BP indicating why the Dunning Lock or Incoming Payment lock was placed on the account.

Figure 10: Create Dunning/Payment Lock

Billing Block

Billing Lock (also referred to as Billing Blocks in S/4HANA) is set on the Contract when the Contract is created and further investigation such as Credit Check returns a warning message or Same Party In is identified during Move In process. Billing Lock can be set to the Contract via Hybris C4C (Figure 11: Create Billing Lock).

The screenshot shows the Hybris C4C interface for managing contracts. The 'Contracts' tab is active, displaying a table with one contract entry. The 'Billing Block' column for this contract is set to 'SCE_D'. Below the contracts table, the 'Locks (0)' table is empty, and a 'Create New Lock' button is highlighted with a red box, indicating the next step in the process.

Figure11: Create Billing Lock

Invoice Lock

An Invoice Lock is set at the Contract Account (Figure 12: create Invoice Lock) when all of the Contracts for the entire Contract Account need to stop the invoice from being created. User will update the Invoice Lock (Tcode CAA2) via Hybris C4C FOP E-CIM-CS-02: Transaction Launcher).

The screenshot shows the Hybris C4C interface for managing invoicing. The 'Invoicing' tab is active, displaying a form with various fields. The 'Lock' checkbox is checked, and a red box highlights the 'Lock' field, indicating the next step in the process.

Figure12: Create Invoice Lock

Process Energy Advice Requests

A BP may request information or assistance to save energy and reduce their energy consumption and/or lower their bill. Requests of this nature can be made by phone, SCE.com, email, web chat, or postal mail. To complete the request, the CCC or BCD representative may simply provide the information the BP requires, or recommend a program to help reduce consumption. For more information about program eligibility and enrollment, refer to BPD [PP-CS-01 Manage Program Strategy, Define & Manage Programs, Manage Catalog](#) and [PP-CS-02 - Program Enrollment and Delivery, De-Enrollment & Decommission Programs](#). An Activity will be created manually by the CCC or BCD representative to document the handling of every request.

Sample Programs:

CARE/FERA

Low income Business Partners (BPs) eligible for the California Alternate Rates for Energy (CARE) program receive discount on their electric bill. The enrollment process for CARE/FERA will be covered in BPD [PP-02: Program Enrollment and Delivery, De-Enrollment and Decommission Programs](#).

Demand Response Program Inquiries

Demand Response Program inquiries can include inquiries for Base-Interruptible Program (BIP), Critical Peak Pricing Program (CPP), Agricultural Pumping Interruptible (API), Summer Discount Program for residential and non-residential (SDP), Capacity Bidding Program (CBP), PTR, DLC, NetGen, Optional Binding Mandatory Curtailment (OBMC), SubTransmission, and Added Facilities. The enrollment process for Demand Response Programs will be covered in BPD [PP-02: Program Enrollment and Delivery, De-Enrollment and Decommission Programs](#).

iEnergy

The representative can access iEnergy via Mashup (*E-CIM-CS-02: Transaction Launcher*) to Hybris C4C when customer inquiries about information on their application and/or rebates.

Activities, Ticket and BPEM Cases

Activity

This functionality replaces CSS notes, requested by and call disposition. The C4C user (e.g., CCC or BCD representative) will log an Activity after resolving the Business Partner (BPs) inquiry and can be used to document what was discussed even if additional follow-up is needed.

Ticket

If the C4C user (e.g., CCC or BCD representative) is unable to resolve the BP inquiry and if the issue needs to be sent to another Front Office group, a Ticket (Figure 13: Create Ticket) in Hybris C4C will have to be created manually. The user will select Ticket Category and select the name of the user or work group that will be working on the Ticket.

The screenshot shows a 'New Ticket' form in a web application. The form is titled 'New Ticket' and has a 'Save and Open' button at the bottom right. The form contains the following fields:

- ID: 448
- Priority: Normal
- Ticket Category: Billing
- Cause Category: (empty)
- Team: HCL Service Org
- Assign To: (empty)
- Subject: Billing Inquiry
- Requested Start Date: 04/08/2018 12:00:00 AM
- Description: Please add a note

Figure13: Create Ticket

Depending on the status of the inquiry investigation, the representative that is working on the Ticket can update the Ticket status to 'Work in Process' or 'Completed' and document the resolution in the Ticket long text (Figure 14: Update Ticket). The statuses of a Ticket are Urgent, Intermediate, Normal and Low. Refer to BPD [CIM-CS-06: Handle Customer Inquiries](#) for more information on Ticket functionality.

Figure14: Update Ticket

A report for Tickets is available in Hybris C4C (Figure 15: Report for Tickets). Using this report, the user is able to see the list of tickets assigned to him/her, total list of ticket created in the system, tickets created last 7 to 365 days. These are standard Hybris C4C selections.

Priority	Type	ID	Subject	Status	Customer	Channel	Reported On	Assigned To	Ins
Normal	Service Req...	291	Billing question	Compl...	Walter Tester	Manual	03/09/2018 3...	Tuan Howe...	
Normal	Service Req...	281	Test	Open	Walter Tester	Manual	03/08/2018 2...	Tuan Howe...	
Immedi...	Service Req...	279	Test	Open	Walter Tester	Manual	03/08/2018 2...	Tuan Howe...	
Normal	Service Req...	254	Customer verification fai...	Open	Walter Tester	Manual	02/15/2018 1...	Tuan Howe...	
Normal	Service Req...	244	Test Moves	Open	Walter Tester	Manual	02/13/2018 2...	Tuan Howe...	
Normal	Service Req...	239	Move In request	Open	Walter Tester	Manual	02/13/2018 2...	Tuan Howe...	
Normal	Employee S...	238	Request for credit	Open		Manual	02/13/2018 2...	Tuan Howe...	
Normal	Employee S...	229	Request for billing infor...	Open		Manual	02/12/2018 1...		

Figure15: Report for Tickets

The report contains filters (Figure 16: Report for Tickets filters) that enables user to filter the list of Tickets based on the filters offered by Hybris C4C such as Assigned To, Status, Reported On, Ticket number, Priority etc..

Priority	Type	ID	Subject	Status	Customer	Channel	Reported On	Assigned To
Normal	Service Req...	291	Billing question	Compl...	Walter Tester	Manual	03/09/2018 3...	Tuan Howe...
Normal	Service Req...	281	Test	Open	Walter Tester	Manual	03/08/2018 2...	Tuan Howe...

Figure16: Report for Tickets filters

BPPEM

If a C4C user (e.g., CCC or BCD representative) is unable to resolve the BP inquiry and if the issue needs to be sent to someone in Back Office group (e.g., RSO, T&D), a BPPEM case will be created manually in Hybris C4C and routed to the appropriate back office workgroup for further handling. A Ticket must be created manually in Hybris C4C prior to creating BPPEM case (Figure 17: BPPEM Case Creation). Upon creating the BPPEM case, the user will also be able to select an appropriate category.

UTILITIES - EXCEPTION MANAGEMENT

Case Category
Business Partner BPPEM

CATEGORY

Service Category
Credit

Incident Category
-

Cause Category
-

Resolution Category
-

Object Category
-

DESCRIPTION

Verification failed.

SERVICE LOCATION

Street
Nuthatch Street

City
Austin

State
Texas

Postal Code
78748

WORK DESCRIPTION

Figure17: BPPEM Case Creation

BPPEM Cases can then be further modified by the Back Office in BDEX/S/4HANA to capture the status, root cause, and results of the bill inquiry. Once they have been resolved, the BPPEM Case status will be updated by the representative manually. In addition, user will update the Ticket status for that BPPEM case.

A summary report of BPPEM Cases is available in S/4HANA by using T-Code /BTR/PP_ASTROLABE to run the report and T-Code /BTR/PP_SUM to display the report. This report provides information regarding BPPEM Status for each type of BPPEM (i.e. new, in process, completed), the number of BPPEMs in each status, created date, and

due date (Figure 18: BDEx Summary Report). To view more details for each BPDM case, the user can double click on the Key Number to open the case.

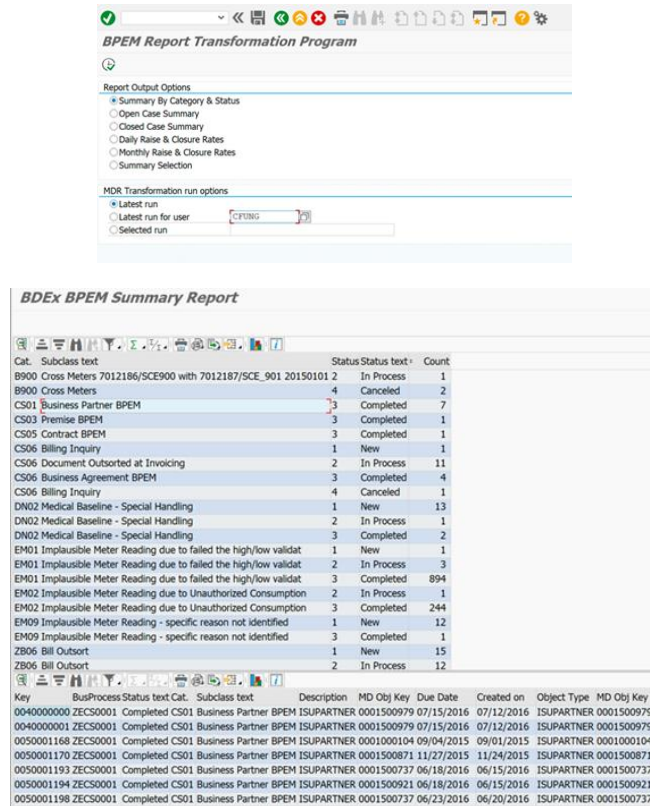


Figure18: BDEx Summary Report

2.1.1 List of Tickets

This section contains the list of tickets that results from Bill Inquiry. A Ticket is created by a Front Office user manually when an issue or request is unresolved and requires further investigation. Tickets can be routed to other C4C users (e.g., a person or a group within Front Office or a Front Office supervisor) manually. A ticket category is defined for a functional area and a ticket category could be used for various scenarios within that functional area. The various scenarios can be differentiated by the user providing the information in the Subject field in the Ticket. The Subject field accepts free form text (Figure 19: Ticket).

The screenshot shows a 'New Ticket' form. The 'Ticket Category' field is highlighted with a red box and contains the text 'Biling'. Other fields include 'ID' (448), 'Priority' (Normal), 'Cause Category', 'Assign To', 'Requested Start Date' (04/08/2018), and 'Time' (12:00:00 AM). There is also a 'Description' field with a placeholder 'Please add a note'.

Figure19: Ticket

Refer to the attached document for a list of Ticket/ Category and responsible department/ person.

No.	Ticket Service Category/Sub Category	Subject Title	Department / Person
Refer to Ticket Worksheet.			

2.1.2 List of BPEM cases

Refer to BPEM case worksheet.

Process / Performance Improvement

- Activities, notes and events are centralized under the Business Partner
- Customer eventually has access to 36 months of historical bills as time goes after system go live

Prerequisites and Inputs

- Business Partner
- Contract Account
- Contract

General Assumptions

- The customer will have up to 12 months of billing history migrated
- There will be users within the CCC and BCD whom have access to the back office. If the action can't be resolved by an agent, a BPEM will be issued to achieve resolution within the CCC/BCD before routing to RSO.

- If a C4C user cannot complete a BI, they will create an Activity and then BPEM to RSO billing.
- All pending Bill Inquiry will need to be extracted from CSS via IT query and manage an excel / Access database until closed or if the volume is low we will re-create them in C4C after go live. This is to mitigate the need for an enhancement to convert them into SAP.
- Extra BI support may be needed to complete as many BIs as possible just before go-live to minimize the effort to manually create open BIs after implementation.
- Change management will be completed with ENAs to ensure they are not sending BIs to the team that should be resolved by them or their lead to reduce the volumes of BIs at go-live.

Business Process Hierarchy

This Business Process Hierarchy provides a listing of all scenarios/business processes, sub-processes, and SAP Transaction codes identified to support this process.

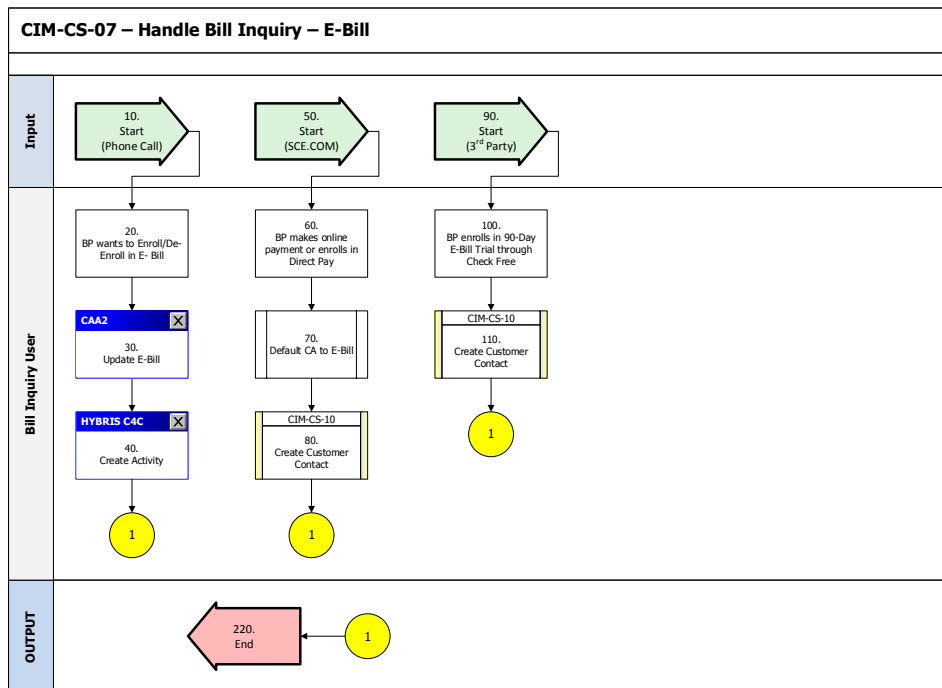
Level 1	Level 2	Level 3	Level 4
FO	Handle Bill Inquiry	Handle Bill Inquiry	Hybris C4C
FO	Handle Bill Inquiry	Handle E-Bill Inquiry	Hybris C4C
FO	Handle Bill Inquiry	Process Energy Advice Requests	Hybris C4C

Process Flows

This BPD will cover 1 process, including Handle Bill Inquiries. Each subsection where the process is detailed will have a process flow diagram for that process and a table detailing the steps that are taken within that process including processes handled outside of SAP.

2.1.3 Handle Bill Inquiry – E-Bill

Process Flow Diagram



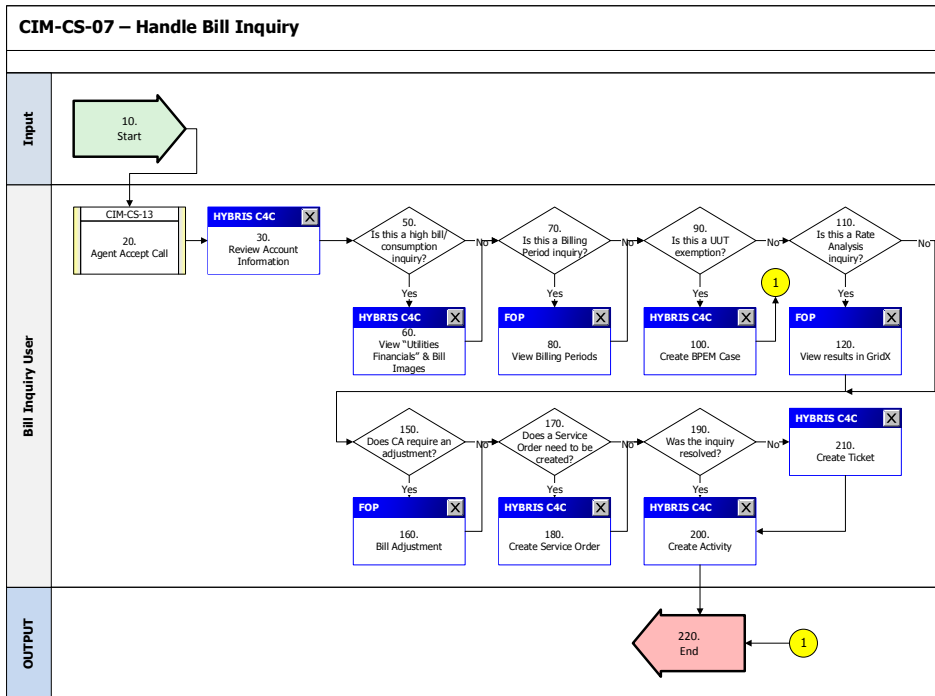
Process Flow Steps

Step	Description of Step	Hybris C4C / T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
10.	Start (Phone Call)	N/A	N/A	
20.	BP wants to Enroll/De-Enroll in E-Bill	Hybris C4C	Bill Inquiry User	
30.	Update E-Bill Update CA with E-Bill	CAA2	Bill Inquiry User	
40.	Create Activity CCC/BCD Representative manually creates an Activity to document enrollment/de-enrollment in E-Bill. Once this is complete, go to Step 120.	Hybris C4C	Bill Inquiry User	
50.	Start (SCE.COM)	N/A	N/A	
60.	BP Makes Online Payment or Enrolls in Direct Pay	N/A	N/A	

Step	Description of Step	Hybris C4C / T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
70.	Default Contract Account to E-Bill When a BP makes an online payment or enrolls in Direct Pay through SCE.COM, the CA is automatically enrolled in E-Bill	S/4HANA	N/A	
80.	Create Customer Contact An Customer Contact is automatically created to document the CA being defaulted to E-Bill	S/4HANA	N/A	
90	Start (3rd Party)	N/A	N/A	
100.	BP Enrolls in 90 E-Bill Trial through Check Free	N/A	N/A	
110.	Create Customer Contact An Customer Contact is automatically created to document the enrollment in E-Bill via Check Free	S/4HANA	N/A	
120.	End	N/A	N/A	

2.1.4 Handle Bill Inquiry

Process Flow Diagram



Process Flow Steps

Step	Description of Step	Hybris C4C / T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
10.	Start	N/A	N/A	
20.	Agent Accepts Call	N/A	Bill Inquiry User	
30.	Review Account Information	Hybris C4C	Bill Inquiry User	
50.	Is this a high bill/consumption inquiry? Decision Point – Is the BP asking questions about the amount due or usage? If yes, go to Step 60. If no, go to Step 70.	Hybris C4C	Bill Inquiry User	
60.	View "Utilities Financials" & Bill Images Review invoice dates, amount due, due dates, open amounts, payments made	Hybris C4C	Bill Inquiry User	

Step	Description of Step	Hybris C4C / T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
70.	Is this a billing period inquiry? Decision Point – Is the BP asking questions about when their next bill is due or if it is currently invoicing? If yes, go to Step 80. If no, go to Step 90.	FOP to Hybris C4C	Bill Inquiry User	
80.	View Billing Periods Review bill period dates for the CA	FOP to Hybris C4C	Bill Inquiry User	
90.	Is this a UUT exemption? Decision Point – Is the BP asking for UUT exemption? If yes, go to Step 100. If no, go to Step 110.	FOP to Hybris C4C	Bill Inquiry User	
100.	Create BPEM Case Create BPEM case for UUT exemption.	FOP to Hybris C4C	Bill Inquiry User	
110.	Is this a rate analysis inquiry? Decision Point – Is the BP asking questions about a rate analysis? If yes, go to Step 120. If no, go to Step 150.	FOP to Hybris C4C	Bill Inquiry User	
120.	View Results in GridX	FOP to Hybris C4C	Bill Inquiry User	
150.	Does CA require a Bill adjustment? Decision Point – Does an adjustment need to be performed to resolve the inquiry? If yes, go to Step 160. If no, go to Step 170.	FOP to Hybris C4C	Bill Inquiry User	
160.	Bill Adjustment	FOP to Hybris C4C	Bill Inquiry User	
170.	Does a Service Order need to be created? Decision Point – Does a Service Order need to be issued to resolve the inquiry? If yes, go to Step 180. If no, go to Step 190.	Hybris C4C	Bill Inquiry User	
180.	Create Service Order	Hybris C4C	Bill Inquiry User	

Step	Description of Step	Hybris C4C / T-Code	Role(s)	If step is relevant for a Control, add Requirement ID
190.	Was the inquiry resolved? Decision Point – was the CCC/BCD representative able to resolve the customer’s billing inquiry? If yes, go to Step 200. If no, go to Step 210.	Hybris C4C	Bill Inquiry User	
200.	Create Activity If the billing inquiry was resolved, the CCC/BCD representative will create an Activity to document the results of what happened on the call	Hybris C4C	Bill Inquiry User	
210.	Create Ticket If the billing inquiry was not resolved and needs to be reviewed further by a different user, the CCC/BCD representative will create a Ticket and route it to the appropriate user for resolution	Hybris C4C	Bill Inquiry User	
220	End	N/A	N/A	

3rd Party and SCE Legacy System Integrations

3 rd Party or SCE Legacy System Integration	Description of Integration and Required Change to 3 rd Party or SCE Legacy System	Estimated Complexity of 3 rd Party or SCE Legacy system change
GridX	Will be made available on user desktop	N/A

Data Elements

The information that will be captured for the below data elements will be defined as the intent of use for this BPD.

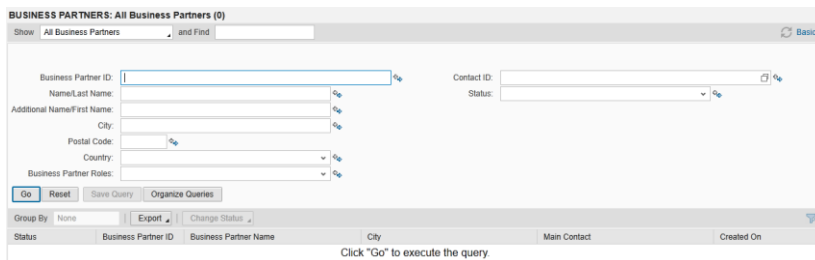
2.1.5 Business Partner Search (Basic)



The following data elements will be used by the Business Partner Search (Basic).

Field	Type	Comments
Find	Required	Business Partner Number

2.1.6 Business Partner Search (Advanced)

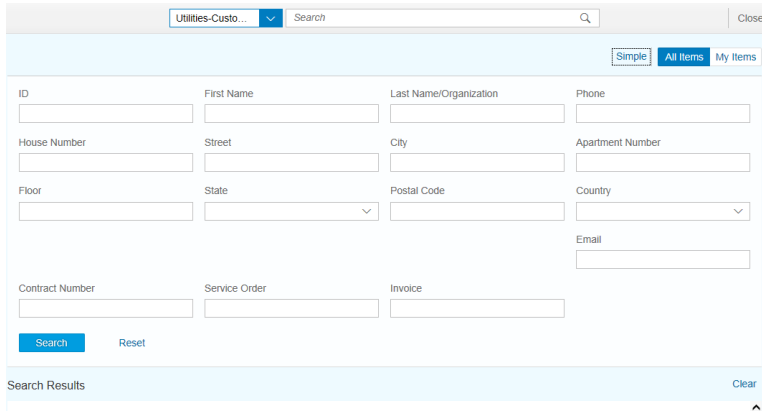


The following data elements will be used by the Business Partner Search (Advanced).

Field	Type	Comments
Business Partner ID	Optional	Business Partner Number
Name/Last Name	Optional	Business Partner's Last Name
Additional Name/First Name	Optional	Business Partner's First Name
City	Optional	City of Business Partner's Address
Postal Code	Optional	Postal Code of Business Partner's Address
Country	Optional	Country of Business Partner's Address
Business Partner Roles	Optional	Example: Sold-to-Party (Refer to BP BPD)
Contact ID	Optional	N/A to SCE
Status	Optional	N/A to SCE

Commented [DT1]: What about Organization name search?

2.1.7 Search Contract Account



The following data elements will be used by the Contract Account Search.

Field	Type	Comments
ID	Optional	Business Partner Number
First Name	Optional	Business Partner's First Name
Last Name/Organization	Optional	Business Partner's Last Name/Organization Name
Phone	Optional	Business Partner's Phone Number
House Number	Optional	House Number of Business Partner's Address
Street	Optional	Street of Business Partner's Address
City	Optional	City of Business Partner's Address
Apartment Number	Optional	Apartment Number of Business Partner's Address (House Number Supplement)
Floor	Optional	N/A to SCE
State	Optional	State of Business Partner's Address
Postal Code	Optional	Postal Code of Business Partner's Address
Country	Optional	Country of Business Partner's Address
Email	Optional	Business Partner's Email Address
Contract Number	Optional	Contract Number
Service Order	Optional	Service Order Number
Invoice	Optional	Invoice Number

2.1.8 Search Premise

Utilities-Premise
Search
Close

Simple
All Items
My Items

Description
ID
Conn. Obj. External ID
House Number

Street
City
Apartment Number
Floor

State
Postal Code
Country
Serial Number

Device Location
Division
PoD External ID

Search
Reset

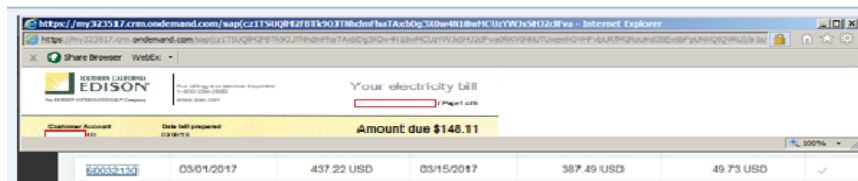
Search Results
Clear

The following data elements will be used by the Premise Search.

Field	Type	Comments
Description	Optional	N/A to SCE
ID	Optional	Premise Number
Conn. Obj. External ID	Optional	Connection Object Number
House Number	Optional	House Number of Business Partner's Address
Street	Optional	Street of Business Partner's Address
City	Optional	City of Business Partner's Address
Apartment Number	Optional	Apartment Number of Business Partner's Address
Floor	Optional	N/A to SCE
State	Optional	State of Business Partner's Address
Postal Code	Optional	Postal Code of Business Partner's Address
Country	Optional	Country of Business Partner's Address
Serial Number	Optional	Meter Serial Number
Device Location	Optional	Device Location Number
Division	Optional	Example: Electric, Gas, Water
PoD External ID	Optional	PoD ID Number

2.1.9 Billing Overview and Bill Image

Invoice Payments Open Items Installment Plan Payment Plan Premise Contracts with Meter Reading						
Invoices (9)						
Invoice No	Invoice Date	Amount Due	Due Date	Open Amount	Payments	Reversed
60032130	03/01/2017	437.22 USD	03/15/2017	387.49 USD	49.73 USD	✓
60032131	03/22/2017	447.85 USD	04/05/2017	396.12 USD	49.73 USD	✓
60032132	03/22/2017	731.39 USD	04/05/2017	294.17 USD	437.22 USD	✓
60032133	03/22/2017	750.57 USD	04/05/2017	313.35 USD	437.22 USD	✓
60032134	03/22/2017	793.42 USD	04/05/2017	793.42 USD	0.00 USD	✓



The following data elements will be used by the Billing Overview and Bill Image.

Field	Type	Comments
Invoice Number	Display	Invoice Number
Invoice Date	Display	Date invoice was generated
Amount Due	Display	Total amount due for invoice
Due Date	Display	Date invoice is due
Open Amount	Display	Remaining amount of invoice still due
Payments	Display	Payments received towards amount due
Reversed	Display	Check box advising if invoice has been reversed

Commented [NM2]: The bill has the tracking for NEM customers. Should this be listed too?

2.1.10 View E-Bill via FOP

Contract Account Info.

ID 5078	Category Utility Consumption Account	External ID 25063602	Name Residential Account
Customer External ID 1008174	Account Class Residential Account	Acc. Detmin. ID Residential Account	Coll. Cont. Acc. -
Coll. Cont. Acc. Ext. -	Dunning Procedure -	Reference No. -	Account in Legacy -
PaymentMethods -			

Financials

Company Code Group -	Company Code 1000	Payment Terms Net Due within 14 Days of Invoicing	Tolerance Group Standard Tolerance Group
BudgetBilling Procedure -	Bill Form -	Alt. Dunning Recipient Ext. -	Alternative Payer Ext. -

Launch FOP

Invoicing

Alt. bill rec.		
Coll. bill acct		
Jurisdct. code	CA00000000	JDC addr. 110 Alcov
ShpCntrl Alt BP		Add
Bill Form	ZPWB_BI013_BILL	Lang
Lock		
Bill consolid.		

The following data elements will be used by the View E-Bill.

Field	Type	Comments
ShpCntrl Alt BP	Optional	Drop-down list containing E-Bill opt-in

2.1.11 Source Systems and Data Volumes

The table below contains the data volumes for bill inquiry received and completed.

Month	BI's Received 2016	BI's Completed 2016	BI's Received 2017	BI's Completed 2017
January	1,341	735	1401	1,146
February	1,246	1,736	1,218	1,333
March	697	936	986	1,160
April	360	516	519	515
May	382	374	566	568
June	478	474	655	654

Month	BI's Received 2016	BI's Completed 2016	BI's Received 2017	BI's Completed 2017
July	744	579	1,081	968
August	1,337	1,238	1,632	1,581
September	961	1,061	1,049	1,090
October	536	701	904	916
November	436	434	601	613
December	749	718	648	684
TOTAL	9,266	9502	11,260	11,228

Cloud for Customer (C4C) Integration

2.1.12 Processes

Bill Inquiries can be handled in Hybris C4C and is available for Front Office agents to display, including search BP, search CA, search Premise, Billing Overview, Bill Image, E-Bill and Service Orders

2.1.13 Hybris C4C Views

- Search BP
- Search CA
- Search Premise
- Billing Overview
- Bill Image
- View E-Bill via FOP
- Service Orders

2.1.14 Activities

An Activity will be created by the CCC or BCD representative manually whenever there is a Bill Inquiry regardless if the bill inquiry is resolved or not. The content that is captured in the Activity long text varies in different Bill Inquiry scenarios. [Refer to Customer Interaction BPD Call Disposition Type & Sub Type](#)

Exception Management Requirements

This section outlines a high level description of each BPEM Case or other user workflow items (such as "Tickets" used by Front Office) used to route work requests between business areas. The companion

document for this section which outlines the detail of each case is documented in APSE on the [Exception Case Template General Document #8](#).

User Exception Scenario / Description	User Group	Initiated by	Work Distribution
Refer to Exception Case worksheet.			

Process Risks and Mitigation

Risk Statement	Mitigation
The option available for rate analysis could be limited due to the features offered by Hybris C4C and GridX. The features of these functionalities are unknown at the moment and SCE is decommissioning of current rate analysis tools.	Utilize worksheet to manually carry out rate analysis.
If a SPOC tool is not available to view MyAccount information and support MyAccount related admin function, there is a risk to need to create more Ticket/BPEM to investigate and resolve customer inquiry.	Increase resources because SCE needs to manually collect the billing and usage information then complete a manual analysis to determine potential recommendation for the customer or identify rebill requirement based on findings.
If the enhancement for billing determinate comparison view is not accepted, the BI team will need to send most inquiry to back office for analysis which will reduce customer satisfaction and increase operation handling and cost.	Increase resources because SCE needs to manually collect the billing and usage information then complete a manual analysis to determine potential recommendation for the customer or identify rebill requirement based on findings.

Key Performance Indicators

The following have been identified as KPI's at Southern California Edison, along with the means of measuring each KPI and the current measurement statistic for each KPI:

Measured KPI	Description	Measurement Method	Current Statistic
Billing Inquiry resolution	Billing Inquiry needs to be resolved	Report	92 % within 16 days. 99 % within 28 days. 100 % within 50 days.

2.1.15 Key Metric Definition

The following have been identified as key metrics at Southern California Edison that require business rule for calculation:

Key Metric	Description	Business Rule	Reference
N/A	N/A	N/A	N/A

3 Controls, Regulatory Rules, and Security

Requirement ID (from RTM)	"As-Is" Control Requirement	"To Be" Risk Statement	"To-Be" Control Activity"	Control Automation (manual, IT dependent- manual, automated)	Relevant technical identifier or RICEFW reference
N/A					

Security Considerations

Role	Transaction Code (T-Code)	T-Code Description	Functional Restriction	Organizational Restriction
Bill Inquiry User	Hybris C4C – Billing Overview	Display Invoices, Payments	Display	No
Bill Inquiry User	Hybris C4C – Open Text	Display Bill Images	Display	No
Bill Inquiry User	Hybris C4C – Service Orders	Create, Change, Display Service Orders	Create, Change, Display	No
Bill Inquiry User	Hybris C4C – Contract Account	Create, Change, Display	Create, Change, Display	No
Bill Inquiry User	Hybris C4C - Invoice	Full Reversal	Create	No
Bill Inquiry User	ES21	Edit Contract	Change	No
Bill Inquiry User	ES22	Display Contract	Display	No
Bill Inquiry User	ES32	Display Installation	Display	No
Bill Inquiry User	ES31	Edit Installation Facts	Edit	No
Bill Inquiry User	CAA2	Edit Contract Account	Change	No
Bill Inquiry User	CAA3	Display Contract Account	Display	No
Bill Inquiry User	EEDM02	Display Interval Data	Display	No
Bill Inquiry User	EA20	Bill Reversal	Create	No
Bill Inquiry User	/BTR/PP_ASTRO LANE	Run BDEx Report	Display	No
Bill Inquiry User	/BTR/PP_SUM	Display BDEx Report	Display	No
Bill Inquiry User	EMMAC2	View, update, complete, cancel, reassign BPEMs	Edit	No

Custom Program/Transaction Codes

Please identify any custom programs/transaction codes that need to be developed for RICEFW objects.

- Is the custom program executed in the batch mode or dialogue mode
- Is the custom program assigned to an authorization group
- What is the custom transaction code for the custom program
- Should there be an authorization check made against any field in the custom program/custom t-code and if so what is the custom/standard authorization object used(For example does it have to be restricted to residential vs commercial customer)
- Does the custom programs/custom t-codes have any critical or sensitive customer information that needs extra security

Custom Table	Authorization Grp Assigned	Custom T-code for the Table	Table Access (display/update)	RICEFW Reference ID
Consolidated Billing view	All users	ZCon_Bill_View		

Custom Table/Transaction Codes

Please identify any custom tables/transaction codes that need to be developed for RICEFW objects.

- Are there any customer sensitive/critical Z tables created
- What is the authorization group assigned to the table
- Is it just a maintenance or display only table
- What is the Z transaction code for the custom table

Custom Table	Authorization Grp Assigned	Custom T-code for the Table	Table Access (display/update)	RICEFW Reference ID

Data Criticality and Cybersecurity Baseline Risks

Specify the Information this BP will process, receive or transmit;	List data elements from section 2.8 here, including volume.			
	Data element	Description	Volume	Risk (high , medium, low)
Specify Access Classification /Special Handling or Regulatory Requirement	<Select all that apply>			
	☒ PII - Personal Identifiable Information / Privacy Law ☐ CEII - Critical Energy Infrastructure Information ☐ FERC SOC (Standards of Conduct) ☐ NERC CIP Confidential ☐ Sarbanes-Oxley (SOX) financial reporting data			

	☐ SCE utility data accessible by affiliate companies (CPUC) ☐ PCI - Payment Card Industry Compliance ☐ Customer Information Service Request (CISR, CISR-DRP) ☐ Other _<Please Specify>_ ☒ Internal ☐ Public									
Please identify which business Group/Role will require access to the data and/or solution:	<Select all that apply> ☒ SCE Customer, Res./Comm. Customer; ☒ SCE OU, Business Function Organization Name: CCC, BCD, Consumer Affairs ☐ Vendor ☐ External business partner(s) & SaaS providers ☐ External on-shore contractors ☐ External off-shore contractors ☐ SCE Affiliate Company Employees ☐ SCE Employees, specify if they have Marketing Function responsibilities (e.g., ES&M) ☐ SCE contract workers ☐ Regulatory Agencies ☐ Others _<Please Specify>_									
How many of the SCE Customer, Employee, Vendors will use or affected by the Business Process?	100; 100 – 1,000;									
How much Information/data would be processed, received or transmitted by the Business Process?	100; 100 – 1,000;									
Number of business transactions in average?										
Legacy System Module (SAP) the Business Process operated on?	SAP Module, Object name; SCE Enterprise System/Application name: 3rd party application integration; <table border="1"> <tr> <th>System</th><th>Module</th><th>Operated By:</th></tr> <tr> <td>CSS</td><td>.....</td><td>MSP</td></tr> <tr> <td></td><td></td><td></td></tr> </table>	System	Module	Operated By:	CSS		MSP			
System	Module	Operated By:								
CSS		MSP								
Targeted System Module	SAP Module, Object name; SCE Enterprise System/Application name: 3rd party application integration; <table border="1"> <tr> <th>System</th><th>Module</th><th>Operated By:</th></tr> <tr> <td></td><td></td><td></td></tr> <tr> <td></td><td></td><td></td></tr> </table>	System	Module	Operated By:						
System	Module	Operated By:								
Risk Rating Please see Appendix for guidance on risk ratings (Very Low, Low, Moderate, High, or Very High) based on business impact should a breach occur in any of the three security	<table border="1"> <tr> <th>Confidential</th><th>Integration</th><th>Availability</th></tr> <tr> <td>H, M, L</td><td>H, M, L</td><td>H, M, L</td></tr> </table>	Confidential	Integration	Availability	H, M, L	H, M, L	H, M, L			
Confidential	Integration	Availability								
H, M, L	H, M, L	H, M, L								
Additional Security Consideration:	Any Cybersecurity impact in addition to SAP system security: - Internet Facing Web Application? Yes/No, Details: - Internet Facing Application Programming Interface? 3rd party Federated Authentication & Authorization, Single Sign On? - Cloud Application (non-C4C)?									

4 Technical Considerations

4.1 RICEF-W Assessment

All of the Forms, Reports, Interfaces, Conversion items, Extensions/Enhancements and Workflows (RICEF-W) that have been identified for this BPD are detailed in the tables below and will represent the totality of all RICEF-W items documented at the time of sign-off of this BPD. Any RICEF-W items that are confirmed via change control after this BPD receives final sign-off, will be managed within Southern California Edison's APSE environment and not within this document.

4.1.1 Reports

Report Name & Description (List Requirements met by this Report)	Complexity	Reference ID

4.1.2 Interfaces

Interface Name & Description (List Requirements met by this Interface)	Complexity	Reference ID
Ability for user to view rate analysis request and result in GridX	Medium	I-CIM-CS-07— Interface to GridX for rate analysis request & result
Ability to view historical weather data to assist with resolving Billing Inquiry	Medium	I-CIM-CS-07— Obtain weather data

4.1.3 Conversion

Conversion Name & Description (List Requirements met by this Conversion)	Complexity	Reference ID

4.1.4 Extension/Enhancements

Extension Name & Description (List Requirements met by this Extension)	Complexity	Reference ID
Ability for agents to sign up customer contract accounts for e-bills Ability for agent to determine which contract accounts are enrolled in e-bill	Medium	E-CIM-CS-07— Enable eBill & Alt Bill Rec field in Front Office screen
Ability for more than one BP (contact) to receive/enroll in paperless billing (eBill) (Ability for agent to link 2 or more BP (contact) to one see.com user id.)	Medium	E-CIM-CS-07— Alt Bill Rec make available in CA view
User has the ability to enroll a customer in eBill when email is available.	Medium	E-CIM-CS-07— Auto default account as paperless during Move-In
Ability for user to send eBill confirmation after enrollment.	Medium	E-CIM-CS-07— Auto send email for eBill enrollment

Extension Name & Description (List Requirements met by this Extension)	Complexity	Reference ID
Ability to compare and present in a consolidated summary view any billed period up to 36 months of history within the customer account. Key elements to be compared: - Period - Amount Due - KWH - KW - KVAR - Usage by Consumption Period (On Peak, Off Peak, etc.) - Daily average kwh usage - Period days - Cost per day	Medium	E-CIM-CS-07 – Bill comparison screen
- Ability to present to the user in a single view the billing/invoice history of the customer - Ability to present in a summary view the following key elements for billed period in process. Key elements to be viewed: - Bill Period - Amount Due - KWH - KW - KVAR - Usage by Consumption Period (On Peak, Off Peak, etc.) - Period days - Daily average kwh usage - Cost per day	Medium	E-CIM-CS-17 - Consolidated Bill View
System will update the Parent task status when a child task is Completed or Cancelled. If there are multiple Child tasks the Parent task will be updated when ALL Child tasks are Complete / Cancelled	Medium	E-CIM-CS-07 – Establish relationship of a parent and child task
- Ability to capture and view UUT (User Utility Tax) indicator - Ability to capture UUT exemption and view UUT exemption status	Medium	E-CIM-CS-07 - User Utility Tax indicator and exception status

4.1.5 Forms

Form Name & Description (List Requirements met by this Form)	Complexity	Reference ID
N/A	N/A	N/A

4.1.6 Workflows

Workflow Name & Description (List Requirements met by this Workflow)	Complexity	Reference ID
N/A	N/A	N/A

4.2 IDEX Considerations

N/A

4.3 Production System Cut-over Considerations

- 12 months of billing history will be migrated over. This will be handled by UB.
- Bill image will not be migrated over, as it is not necessary, they will be available in Open Text
- Bill image generated by OpenText starting 2018 will be made available in CSRP system during cut-over.
- SCE will need IT to extract a list of all active BI related cases prior to go-live so they can be managed outside of SAP until closed or manually created after go-live.

4.4 Other Considerations

N/A

5 Change Management

5.1 Change Impacts

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
In SAP, RSO will be able to complete a billing reversal without having to cancel any future dated move in requests. Currently to complete any billing reversals, RSO needs to cancel the turn on request so that they can make billing corrections and then re-issue the turn on request. The time needed to complete the requests can be minimized since the user will not need to re-issue any other orders.	Process (150)	CS-CSOD-RSO All	Low	Low
Front office users will access back office screens using the Front Office Process (FOP) launcher. It is a pending enhancement. The launcher creates an opportunity for users to view data and issue transactions that is not available in the front office UI (eg. installation facts, interval usage, utility tax exemption, joint billing etc.) and minimizes the need for a hand off to the back office. This creates a single sign on to the back office system from the front office. Currently CCC and BCD use one major system for most transactions, CWO and Legacy CRM respectively. Users will need to know when to use the FOP Launcher for transactions and data elements.	Culture Shift;Benefit/Improvement;Process;Decision Making (815)	CS-BCD-All;CS-CP&S-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs	High	High

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
In the front office UI, a ticket must be issued to create a BPEM case to the back office. A BPEM can not be accessed without issuing the ticket. There will be multiple steps related to creating the request for the case. Currently users issue a basic work request to send a request to the back office. Training will be required on the fields that need to be completed within the ticket and BPEM case to ensure accuracy.	Process (816)	CS-BCD-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs	High	High
The ticket listing functionality creates a flexible view for users that will work on completing ticket requests. Users will have the ability to customize views (eg view all, your own, by customer etc) with multiple search functions and the ability to filter and sort. Reports can be run on tickets by authorized users to review completion rates and due dates to ensure service levels are met. Currently users use email or a queue functionality to view workload especially when it is communication via front office user to front office user.	Benefit/Improvement;Culture Shift;Process (817)	CS-BCD-All;CS-CP&S-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs	High	Medium
Users will have the ability to add another BP to receive the bill via the back office. Front office will use the FOP launcher to access the back office to issue the request. This functionality is currently not available in CSS/CWO/Legacy CRM today. Customer Impact: Customers have the flexibility to request another BP receive a copy of their bill. For instance, parents request copies of bills for their adult children or elderly dependents to assist them with their bills.	Benefit/Improvement;Decision Making;Process (818)	CS-BCD-All;CS-CP&S-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs;CS-CSOD-RSO All	Medium	Medium

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
Billing Inquiry (BI) tool functionality will not be available in front office. Weather data and appliance mix calculation will not be available in SAP. New processes or job aids will need to be created to assist users with completing billing inquiries. Customer Impact: Indirect impact to customers. It may increase the amount of billing inquiries that need to go to level 2 causing a longer turn around time for customers instead of first call resolution.	Culture Shift;Decision Making;Process (819)	CS-CSOD-CCC All;CS-CSOD-Consumer Affairs	High	High
Business Customer Division (BCD) Account Managers will have the ability to enroll/de-enroll some customers to e-bill without a hand off to BCD sce.com COE. Large corporations may still need to follow the current process to validate corporate structures. Currently all enrollments are sent to a specialized group in BCD (sce.com COE). As a result, we achieve first call resolution, however, Account Managers will need to complete the end to end process.	Benefit/Improve ment; Process; Job Roles (290)	CS-BCD-All	Medium	Medium
Business Customer Division (BCD) will have the ability to create a BPEM via a ticket to different departments for multiple requests such as delayed billing, Direct Install requests etc. Currently requests are centralized and sent to COE for review and then sent to other organizations. SAP process minimizes hand offs and provides a faster resolution for the customer with minimal processing time.	Process; Job Roles; Benefit/Improve ment (300)	CS-BCD-All	Medium	Medium

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
Billing Inquiry may not have a listing view that is easily accessible for certain information that will assist the CCC with bill questions, such as daily average and KW demand. May have to drill down or click to another screen to find the information - requirements have been added for a list view screen. Enhancement is pending.	Decision Making; Process (363)	CS-CSOD-CCC All; CS-BCD-All; CS-CSOD-Consumer Affairs	Medium	Medium
Payment transfer from 1 account to another will take 2 steps in SAP. The agent will need to move the payment from one account to a clarification work list and then move from the work list to another account. Current process is one step based on selecting the payment. This can add more processing time for the agents that will need to move payments.	Decision Making; Process	CS-CSOD-CCC All	Low	Medium
Front office users will not have visibility to BPEMs that are created in the back office or the status of BPEMs that they create via C4C. Currently users have visibility to all basic work requests issued in CSS/CWO. For some teams, a workaround will be to use BDEX to view BPEM info (e.g., BI team) and they can review notes (Activity Records and BP Contact records) in C4C to view the full picture of the account.	Process, Decision (827)	CCC, GC Services, BCD, Consumer Affairs, CP&S	High	High

Change Element Short Description	Change Impact Category	Audience Group(s)	Impact	Business Effort
The BI team will need to use three systems to receive, view and complete their work including C4C, S/4 Hana and BDEX. Today, the BI Team manages their work in CSS and an ACCESS DB. There will be additional steps needed to ensure that they get the full picture of the account including any pending events (eg BPDM cases, service order, rebills etc), notes and usage data.	Process (828)	CS-CSOD-CCC-BI	Med	Med
Users will not be able to view any change logs on tickets. The view will only be based on the last user that worked on the ticket. Users will not be able to view who made changes to the ticket request or what updates were made to the tickets. This can be an issue when researching the account for escalations. Currently users have a view log that shows changes made to requests in CSS/CWO.	Decision Making;Process;Regulatory/Legal (844)	CS-BCD-All;CS-CP&S-All;CS-CSOD-CCC All;CS-CSOD-Consumer Affairs;CS-CSOD-RSO All	Medium	Medium

Below is a description of each:

Change Element Short Description: Describes the ‘As-Is’ scenario and what is changing.

Change Impact Category: Determined by asking the questions below.

1. **Staffing Levels** - Is the work effort increasing or decreasing (staffing levels), and how (e.g., task automation, new process/tasks)?
2. **Job Roles** - Will the responsibilities, authority, or required skills for a job or job roles change? Will people be reporting to new leaders? Will managers’ responsibilities change?
3. **Compensation / Performance** - Will the changes have such an impact on roles of an individual that merits a change in how they are compensated or how their performance will be evaluated? What behaviors are rewarded and recognized?
4. **Organization Structure** - Is the ownership of the Process or Data changing (think departments, jobs, or roles, not individuals)?
5. **Decision Making** - Will any of the changes impact how decisions are made, by or between, business units / departments? Who makes them, how they determine the criteria for making a change?
6. **Culture Shift** - Will organizational value systems impact the way change happens or be impacted by the change? What are the underlying values of the organization that will directly influence the way change will be accepted and how much work will ultimately be required to ensure successful outcomes for the business?

7. **Location / Facilities** - Will the change impact the physical location where they process is being implemented or require changes to the facilities to enable the process to be effective?
8. **Process** - Any change to the current process (e.g. manual process becomes automatic, or vice versa)? Are there new processes being introduced or an old process being decommissioned? Is there a requirement to change an existing business rule or business policy? Will new business rule or policy needs to be introduced and enforced or complied to?
9. **Regulatory/ Legal** - Are there any legal / regulatory requirements that might be impacted
10. **Technology** - Are other applications impacted (e.g., going away, becoming read-only), or are there new requirements (e.g., new SW / HW, new version)
11. **Benefit/Improvement** - Will there be an improvement or benefit to any of the stakeholders because of this change. (Be sure to include any pain points that are being alleviated)
12. **Other** - Are there emotional implications stemming from any of these changes (think of buy-in issues or fears)

Audience Group(s): These are the affected departments, divisions etc...

Impact: The following were used to determine the impact:

Level	
HIGH	- BUSINESS OUTCOMES: Drives change to cost, efficiency, customers. - JOB: More than 40% of individual's job will change and / or additional skills are required with significant changes to Job Documents. Examples include: - Shift from manually intensive role to using new tools and systems - Employee is going to be redeployed - High level of "technical or functional" changes needed for a particular group - Significant changes to culture & attitudes to get job done - Big shifts in accountability or reporting (autonomy, responsibility, decision making, etc.) - RELATIONSHIPS: High level of "relational" changes for a particular group (e.g. reporting relationships, effective working relationships with customers, teams, and stakeholders etc.) - POLICIES/PROCEDURES: Processes, procedures, and/ or policies are not in place today and need to be defined. - REGULATORY REQUIREMENT: drive changes to the way we comply
Level	
MEDIUM	- BUSINESS OUTCOMES: Limited change to cost, efficiency, customers. - JOB: 20-40% of individual's job will change and / or additional skills are required with significant changes to Job Documents. Examples include: - Employee job title or function is modified, resulting in minor to moderate changes to roles and responsibilities. - There will be some changes to the culture & attitudes - Shift from moderate technology use to using new tools and systems - Moderate level of "technical or functional" changes needed for a particular group - Moderate shifts in accountability or reporting (autonomy, responsibility, decision making, etc.) - RELATIONSHIPS: Moderate level of "relational" changes for a particular group (e.g. reporting relationships) - POLICIES/PROCEDURES: Processes, procedures, and/or policies are inconsistent today and not defined or documented in many cases - REGULATORY REQUIREMENT: No impact.

Level	
LOW 	• BUSINESS OUTCOMES: No impact. • JOB: Less than 20% of individual's job will change and / or additional skills are required with significant changes to Job Documents. Examples include: • Employee job title or function is slightly modified with few, if any, changes to roles and responsibilities • There will be little change to the culture & attitudes • Shift from using some technology to new systems & tools • Light level of "technical or functional" changes needed for a particular group • Small shifts in accountability or reporting (autonomy, responsibility, decision making, etc.) • RELATIONSHIPS: No impact or light level of "relational" changes for a particular group (e.g. reporting relationships) • POLICIES/PROCEDURES: Processes, procedures, and/or policies are consistent, but not well documented, and will need to be documented/ Other minor changes to be made. • REGULATORY REQUIREMENT: No impact.

Business Effort: Initial assessment of the impact to the business High, Medium, Low

6 Requirements

Requirement #	APSE Requirement #	Requirement Description

7 Tickets

The attached document contains the list of Tickets that identified in this BPD.

8. Appendix



eBill Visio v1.0.vsd



Bill Inquiry Visio
V1.0.vsd